

B.TECH/EE/ECE/IT/ME/CE/ODD/1st SEM/R_21//2022-2023

YEAR: 2023

CHEMISTRY-I

CH 101

TIME ALLOTTED: 3 HOURS

FULL MARKS: 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable

GROUP – A

(Multiple Choice Type Questions)

1. Answer any **ten** from the following, choosing the correct alternative of each question: **10×1=10**

SL	Question	Marks
(i)	The possible set of quantum no. for the unpaired electron of sodium is a) $n=2, l=1, m=0$ b) $n=2, l=1, m=1$ c) $n=3, l=1, m=1$ d) $n=3, l=0, m=0$	1
(ii)	Racemization occurs in which of the following? a) E1 reaction b) SN2 reaction c) SN1 reaction d) free radical halogenations	1
(iii)	IR spectra normally carried on a) Liquid sample b) Gas samples c) Solid samples d) All	1
(iv)	How many spectral emission lines are conventionally obtained in the emission spectra of H atom a) 5 b) 2 c) 4 d) 3	1
(v)	Paracetamol is used to treat a) orthopedic ailments b) stomach ailments c) pain and fever d) lungs infection	1
(vi)	Ion exchange method is used a) To remove permanent hardness of water b) To protect metal from corrosion c) To measure alkalinity of water d) To measure chloride ion in water	1

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| (vii) For ideal gases, compressibility factor (Z) is | 1 |
| a) < 1 | |
| b) $= 1$ | |
| c) $= 0$ | |
| d) > 1 | |
| (viii) The most stable carbonation among the following is | 1 |
| a) CH_3^+ | |
| b) $(\text{CH}_3)_2\text{CH}^+$ | |
| c) $(\text{CH}_3)_3\text{C}^+$ | |
| d) PhCH_2^+ | |
| (ix) Phenol formaldehyde resin is an example of | 1 |
| a) Addition polymer | |
| b) Isotactic polymer | |
| c) Thermoplastic polymer | |
| d) Thermosetting polymer | |
| (x) The role of salt bridge is | 1 |
| a) to keep e.m.f of the cell positive | |
| b) to keep e.m.f of the cell negative | |
| c) to maintain electrical neutrality of the solution in two half cells | |
| d) none of the above. | |
| (xi) The quantity of electricity needed to liberate one gm equivalent of an element is | 1 |
| a) 1 ampere | |
| b) 96500 amperes | |
| c) 96500 coulomb | |
| d) 96500 faradays | |
| (xii) Which of the following equation is incorrect? | 1 |
| a) $H = G + TS$ | |
| b) $G = H + TS$ | |
| c) $A = U - TS$ | |
| d) $H = U + PV$ | |

GROUP – B

(Short Answer Type Questions)

(Answer any three of the following) 3 x 5 = 15

- | SL | Question | Marks |
|----|--|-------|
| 2. | (i) Draw the resonating structures of PhOH | 2 |
| | (ii) Write Vander Walls's equation of state for n moles of a real gas. Explain the terms involved. | 3 |
| 3. | Write a short note on Biodegradable Polymers. | 5 |
| 4. | 56 grams of a gas was initially at 50 atm and 250C. It was allowed to expand isothermally against a constant external pressure of one atm. Calculate ΔU , ΔQ and W, assuming the gas to behave ideally. (molar mass of the gas = 28) | 5 |

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| 5. | (i) Write down the Schrödinger wave equation in its standard form and define each term in it. | 2 |
| | (ii) Write in brief the physical significance of entropy. | 3 |
| 6. | (i) 1 mole of a gas occupying 5 litre is expanded against a constant external pressure of 1 atm to a volume of 11 litre. Calculate work. | 2 |
| | (ii) Predict whether in the following reactions entropy change is positive or negative | 3 |
| | $\text{NH}_4\text{Cl(s)} \rightarrow \text{NH}_3\text{(gas)} + \text{HCl}$ | |
| | $\text{C graphite} + \text{CO}_2\text{(g)} \rightarrow 2\text{CO(g)}$ | |
| | Freezing of water | |

GROUP – C
(Long Answer Type Questions)

(Answer any three of the following) 3 x 15 = 45

- | SL | Question | Marks |
|----|--|-------|
| 7. | (i) Write the postulates of Bohr atomic theory. | 3 |
| | (ii) Define electron affinity. How does electro negativity depend on oxidation state and hybridization state of atom? | 4 |
| | (iii) What is the radius of 3rd Bohr orbit of hydrogen atom? (Given: $h = 6.626 \times 10^{-27}$ erg.s, mass of electron = 9.1×10^{-28} g, charge of electron = 4.8×10^{-10} esu) | 3 |
| | (iv) If uncertainty in position and velocity are equal, then find the uncertainty on momentum. | 2 |
| | (v) Write the possible set of quantum number for a valence electron in ground state atom of phosphorus ($Z=15$). | 3 |
| 8. | (i) How are Aspirin and Paracetamol synthesized? | 4 |
| | (ii) How can Ion Exchange process be applied to produce de-mineralized water? | 6 |
| | (iii) Differentiate between Thermoplastics and Thermosets. | 5 |
| 9. | (i) A cell reaction is : $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$, write the correct cell notation. Mention the cathode and anode. | 2+1 |
| | (ii) For the redox couples $\text{Zn}^{2+} \text{Zn (s)}$ and $\text{Cu}^{2+} \text{Cu (s)}$ the standard reduction potentials are -0.76 V and + 0.34 V respectively. If the couples are combined then which metal will be oxidized, explain. | 2 |
| | (iii) Write three applications of Nernst Equation. | 3 |
| | (iv) Calculate ΔG° for the cell reaction between $\text{Zn} + \text{FeSO}_4 \rightarrow \text{Fe} + \text{ZnSO}_4$
$E^\circ_{(\text{Zn}^{2+}/\text{Zn})} = -0.76\text{V}$, $E^\circ_{(\text{Fe}^{2+}/\text{Fe})} = -0.44\text{V}$. Predict whether the reaction is spontaneous or not. | 3+1 |
| | (v) Why real gases deviate ideal gas laws? | 3 |

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| 10. | (i) | What are the transitions possible in a general organic compound containing a C=O group? Arrange them in order of increasing energy. | 2 |
| | (ii) | What is chromophore and auxochrome? Give examples. | 3 |
| | (iii) | Define electromagnetic spectrum. What is the wavelength range of the visible region? | 2 |
| | (iv) | Write a note on SN1 reaction. | 5 |
| | (v) | What is IR spectroscopy? What are its most significant applications? | 1+2 |
| 11. | (i) | What is tacticity? Distinguish between Isotactic, Syndiotactic and Atactic Polymers with examples. | 2+6 |
| | (ii) | Describe the process of hardness removal by Permutit process. | 4 |
| | (iii) | Differentiate between Thermoplastics and Thermosets. | 3 |